

L^AT_EX Mathematics Examples

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1 Delimiters

See how the delimiters are of reasonable size in these examples

$$(a+b)\left[1-\frac{b}{a+b}\right]=a\,,$$

$$\sqrt{|xy|}\leq\left|\frac{x+y}{2}\right|\,,$$

even when there is no matching delimiter

$$\int_a^b u\frac{d^2v}{dx^2}dx=u\frac{dv}{dx}\Big|_a^b-\int_a^b\frac{du}{dx}\frac{dv}{dx}dx\,.$$

2 Spacing

Integrals often need a bit of help with their spacing as in

$$\iint xy^2 dx dy = \frac{1}{6}x^2y^3,$$

whereas vector problems often lead to statements such as

$$u = \frac{-y}{x^2 + y^2}, \quad v = \frac{x}{x^2 + y^2}, \quad \text{and} \quad w = 0.$$

3 Arrays

Arrays of mathematics are typeset using a tabular like environment as in

$$\begin{bmatrix} 1 & x & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 1 + xy \\ y - 1 \end{bmatrix},$$

or in a case statement such as

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Many arrays have lots of dots all over the place as in

$$\begin{bmatrix} -2 & 1 & 0 & 0 & \cdots & 0 \\ 1 & -2 & 1 & 0 & \cdots & 0 \\ 0 & 1 & -2 & 1 & \cdots & 0 \\ 0 & 0 & 1 & -2 & \ddots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \ddots & 1 \\ 0 & 0 & 0 & \cdots & 1 & -2 \end{bmatrix}$$

4 Equation arrays

In the flow of a fluid film we may report

$$u_\alpha = \epsilon^2 \kappa_{xxx} \left(y - \frac{1}{2}y^2 \right), \tag{1}$$

$$v = \epsilon^3 \kappa_{xxx} y, \tag{2}$$

$$p = \epsilon \kappa_{xx}. \tag{3}$$

Alternatively, the curl of a vector field (u, v, w) may be written with only one equation number:

$$\begin{aligned}\omega_1 &= \frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}, \\ \omega_2 &= \frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}, \\ \omega_3 &= \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}.\end{aligned}\tag{4}$$

Whereas a derivation may look like

$$\begin{aligned}(p \wedge q) \vee (p \wedge \neg q) &= p \wedge (q \vee \neg q) && \text{by distributive law} \\ &= p \wedge T && \text{by excluded middle} \\ &= p && \text{by identity}\end{aligned}$$

5 Functions

Observe that trigonometric and other elementary functions are typeset properly, even to the extent of providing a thin space if followed by a single letter argument:

$$\exp(i\theta) = \cos \theta + i \sin \theta, \quad \sinh(\log x) = \frac{1}{2} \left(x - \frac{1}{x} \right).$$

With sub- and super-scripts placed properly on more complicated functions,

$$\lim_{q \rightarrow \infty} \|f(x)\|_q = \max_x |f(x)|,$$

and large operators, such as integrals and

$$\begin{aligned}e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} \quad \text{where } n! = \prod_{i=1}^n i, \\ \overline{U_\alpha} &= \bigcap_{\alpha} U_\alpha.\end{aligned}$$

Although in inline mathematics the scripts are placed to the side in order to conserve vertical space, as in $1/(1-x) = \sum_{n=0}^{\infty} x^n$.

6 Accents

Mathematical accents are performed by a short command with one argument, such as

$$\tilde{f}(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx,$$

or

$$\dot{\vec{\omega}} = \vec{r} \times \vec{I}.$$

7 Command definition

The Airy function, $\text{Ai}(x)$, may be incorrectly defined as this integral

$$\text{Ai}(x) = \int \exp(s^3 + isx) ds.$$

This vector identity will serve nicely to illustrate two of the new commands:

$$\nabla \times \mathbf{q} = \mathbf{i} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) + \mathbf{j} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) + \mathbf{k} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right).$$